Renewable Electricity Trends

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Energy Data Analysis

Project Report

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Overview

This project follows renewable electricity in India, Germany, and Brazil from 2000 through 2024. A Python notebook prepares the Our World in Data Energy dataset and compares renewable share, generation, and source mix (Our World in Data, 2026). Germany records the largest rise in renewable share, while Brazil stays highest across the study period.

Dataset Description

The September 10, 2026 release contains 19,494 rows and 130 columns, with annual country and regional observations from 1700 through 2025. The complete CSV is stored beside the notebook, and the analysis selects 75 records covering 25 years for each country. Eight numeric variables cover renewable share, renewable and total generation, and shares from hydro, wind, solar, bioenergy, and other renewables excluding bioenergy.

Shares use total electricity generation as their denominator, and generation is measured in terawatt hours (TWh). The electricity fields combine records from Ember, the Energy Institute, and historical sources, with processing by Our World in Data (2026). Values for 2025 are available, but the study keeps 2000 through 2024 as its defined 25 year scope.

Workflow Description

The workflow loads the CSV and displays its first rows. Two functions standardize column names and prepare the country and year records. A cleaning audit and quality checks confirm complete annual coverage and consistent measures. An exploratory function compares the endpoints, annual peaks, and generation ratios, followed by three charts and a written summary.

Code, outputs, and explanations stay together in the notebook. This follows Danchev's (2022) use of notebooks to connect data inputs, calculations, figures, and narrative in a transparent workflow. The local source file and package versions recorded with pip freeze make the run easier to reproduce.

Key Decisions and Assumptions

Column names and numeric types are standardized so calculations use consistent fields and values. Cleaning removes duplicate analysis rows only when all selected fields agree. Conflicting records or invalid numbers stop the workflow, and existing missing values remain missing. The selected data need no value replacements or duplicate removal, which the audit records as zero changes.

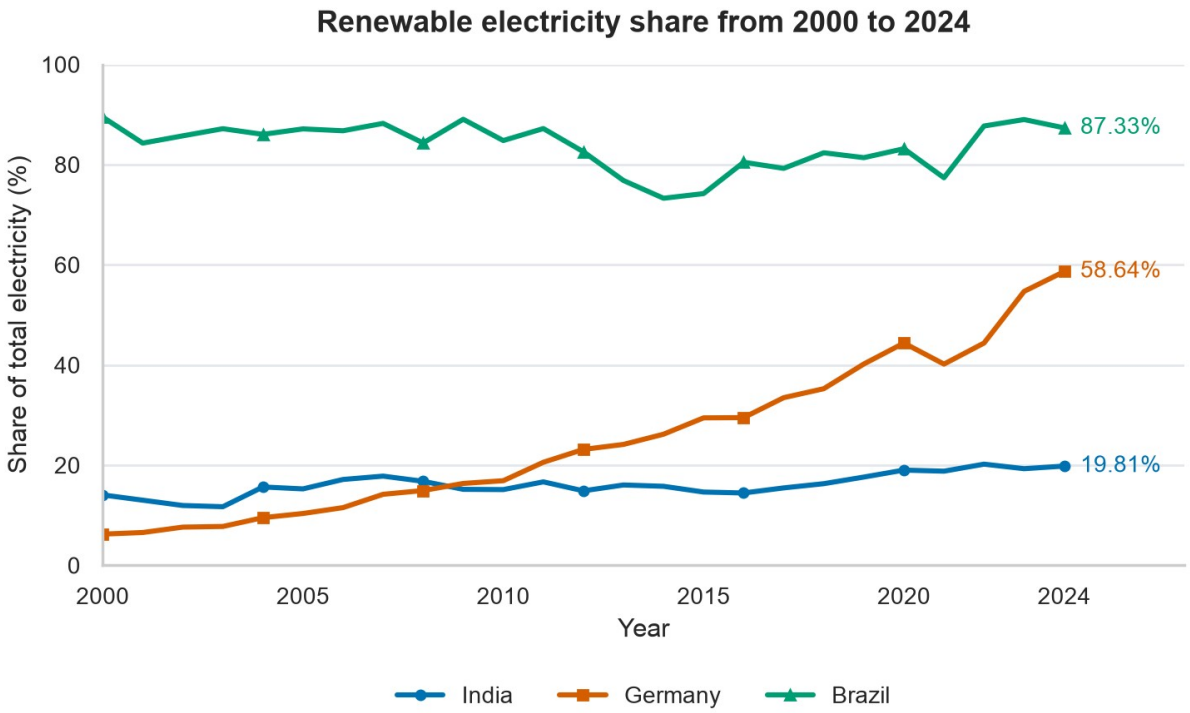
The analysis compares annual paths as well as endpoints, because an endpoint can hide a peak or recovery. Share changes use percentage points, and generation ratios divide 2024 output by 2000 output. Figure 1 compares renewable share on a zero to 100 percent scale, Figure 2 compares generation from a zero baseline, and Figure 3 separates the renewable sources using the same total electricity denominator.

Results and Interpretation

Figure 1 shows Germany's renewable share rising from 6.22% to 58.64%, a gain of 52.42 percentage points. India's share rises from 14.05% to 19.81%, below its study peak of 20.19% in 2022. Brazil stays highest throughout the period, falls to 73.28% in 2014, and recovers to 87.33% in 2024. Its final share is 2.16 points below its 2000 value.

Figure 1

Renewable electricity share from 2000 to 2024

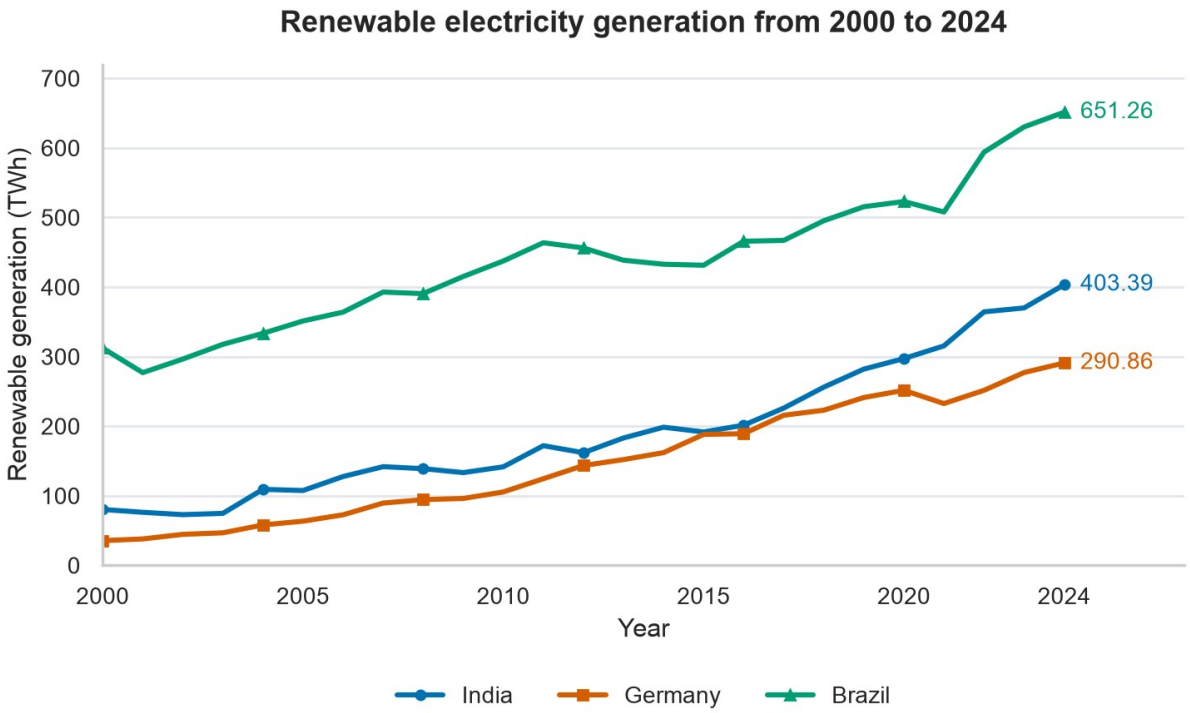


Note. Values show renewable electricity as a share of total electricity generation.

Figure 2 shows renewable generation rising from 312.26 to 651.26 TWh in Brazil, from 80.27 to 403.39 TWh in India, and from 35.41 to 290.86 TWh in Germany. The exploratory table explains the share changes: India's renewable output reaches 5.03 times its starting level, while total output reaches 3.56 times. In Brazil, renewable output reaches 2.09 times its starting level, while total output grows slightly faster at 2.14 times.

Figure 2

Renewable electricity generation from 2000 to 2024

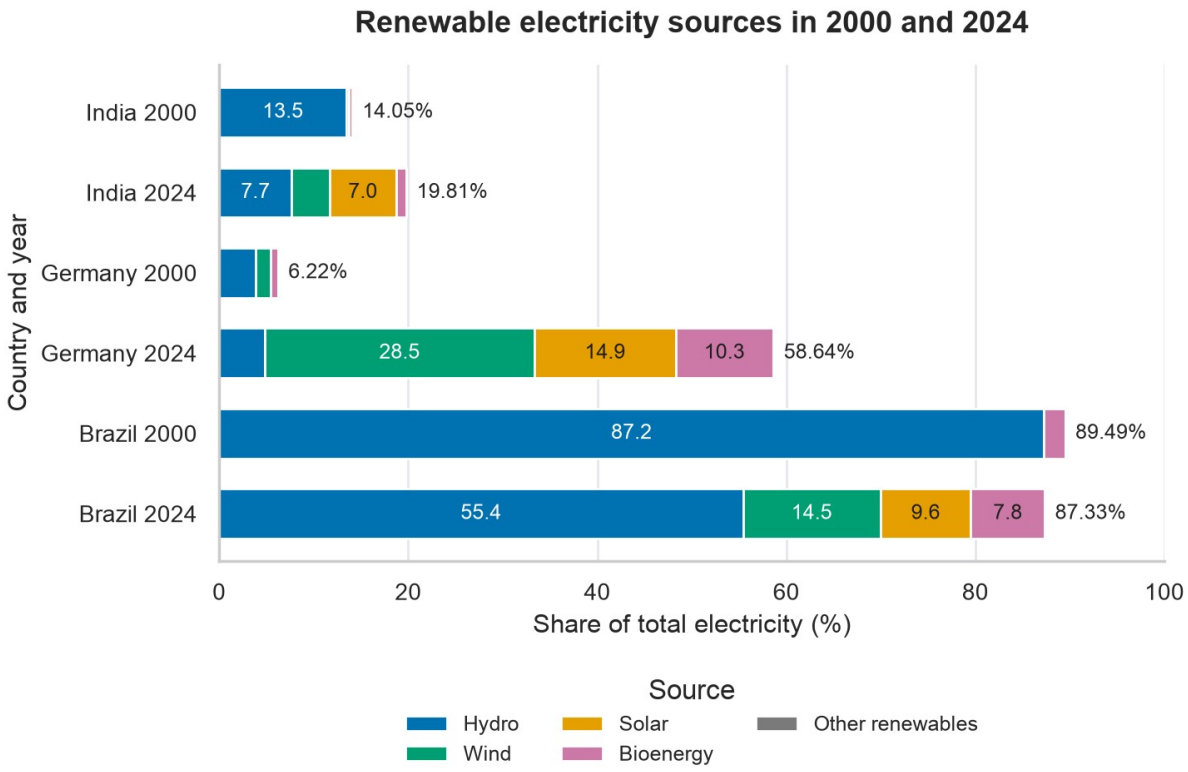


Note. Annual renewable electricity generation is reported in terawatt hours.

Figure 3 shows Brazil's hydro share falling from 87.24% to 55.41% of total electricity, while wind and solar reach 14.55% and 9.56% in 2024. Wind is Germany's largest renewable source at 28.55%, followed by solar at 14.95%. In India, hydro at 7.69% and solar at 7.03% account for similar shares. Wind and solar contribute larger shares in 2024 than in 2000 in all three countries.

Figure 3

Renewable electricity sources in 2000 and 2024



Note. Component shares use total electricity generation as the denominator.

Responsible Practice (Bias and Data Quality)

All 75 study records are complete for the eight selected measures, so no values are filled or estimated. Removing years with missing data could favor countries with better reporting, and replacing unknown values with zero could lower a trend artificially. Missing data methods depend on assumptions about why values are absent (Schafer & Graham, 2002). A later release with gaps would require checking the source notes before choosing a treatment.

The comparison assumes that country definitions and electricity measures are comparable within this release. Multiple providers and historical series can introduce differences that the numeric checks alone cannot resolve. National totals hide regional variation, and three selected countries cannot represent global trends. The figures describe the series without establishing causes or measuring cost, grid reliability, access, or emissions.

Reproducibility

The final run uses Python 3.14.6 and the package versions in requirements.txt, created with pip freeze. The README explains environment setup and how to run data_workflow.ipynb from a restarted kernel with owid_energy.csv beside it. The notebook was also executed in a fresh environment installed from that package file. Multiple commits record progress, and the development branch holds revisions before they are merged into main.

Sources and Citations

Earlier G20 research reports country differences and a growing role for wind and solar (Sokulski et al., 2022). That study uses IEA data from 1990 through 2020 and provides context for comparing the source mix. All numerical results and figures here are calculated from the included Our World in Data release for 2000 through 2024.

References

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, *5*(56), 156. https://doi.org/10.21105/jose.00156

Our World in Data. (2026, September 10). *Energy dataset* [Data set]. https://catalog.ourworldindata.org/garden/energy/2026-09-10/owid_energy/owid_energy.csv

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Sokulski, C. C., Barros, M. V., Salvador, R., Broday, E. E., & de Francisco, A. C. (2022). Trends in renewable electricity generation in the G20 countries: An analysis of the 1990–2020 period. *Sustainability*, *14*(4), Article 2084. https://doi.org/10.3390/su14042084